LSTM and GRU comparison results report

Multi-pollutant forecasting | 4 November 2026

Dataset: supplied combined hourly CSV with 11 pollutants and 14,592 rows.

Training and validation use 2025. Testing uses January to August 2026.

Daily: hourly forecasts averaged to a day versus a direct daily forecast.

Weekly: daily forecasts averaged to a week versus a direct weekly forecast.

Monthly: weekly blocks averaged to a month versus a direct monthly forecast.

Error = predicted - actual. Each prediction row includes this exact signed error.

Find every prediction and its error in each model folder: results/predictions_2026.csv.

MAE is the average absolute error. RMSE gives larger errors more weight.

Lower MAE and RMSE are better. Compare values within the same pollutant.

Units: CO is mg/m3; NOx is ppb; other pollutants use source micrograms/m3.

Each actual average needs 75% hourly coverage. Missing actuals are not scored.

Forecasts use measurements available before the period starts.

Later 2026 forecasts can use earlier observed 2026 data, without retraining.

Monthly findings are exploratory: only nine direct training windows and eight test months.

One seeded run does not establish superiority. AQI classification is still pending.

Network: one recurrent layer with 32 hidden units; Adam at 0.001; seed 42.

Validate on October-December 2025; select epochs, then refit on eligible 2025 data.

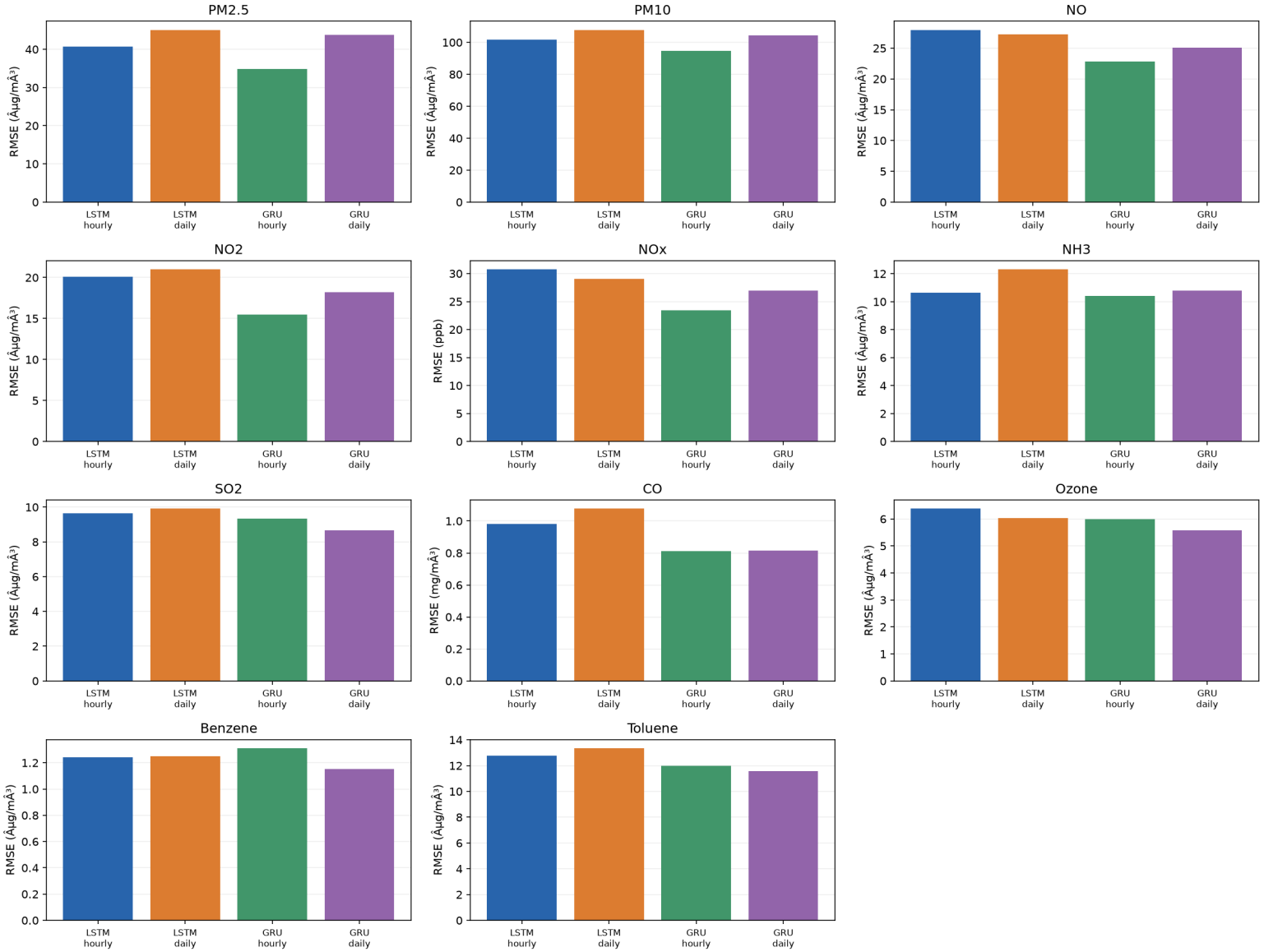
Missing past inputs fill forward or use training medians; scaling uses training data only.

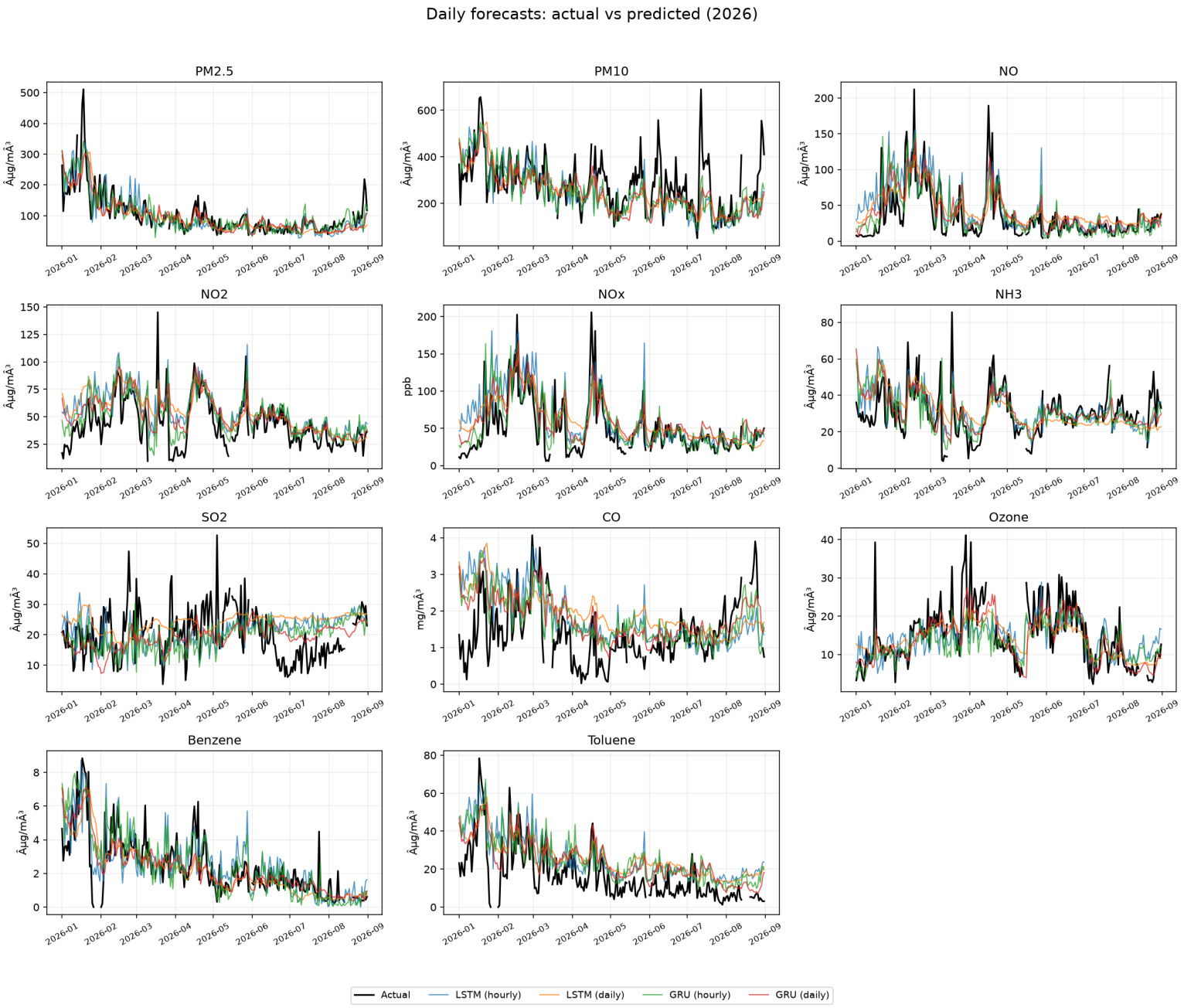
Daily errors

MAE and RMSE in original units. Lower is better. N is the number of scored periods.

Pollutant	N	GRU daily MAE	GRU daily RMSE	GRU hourly MAE	GRU hourly RMSE	LSTM daily MAE	LSTM daily RMSE	LSTM hourly MAE	LSTM hourly RMSE
Benzene	229	0.78	1.15	0.95	1.31	0.81	1.25	0.93	1.24
CO	228	0.63	0.81	0.65	0.81	0.90	1.08	0.79	0.98
NH3	229	7.76	10.80	7.40	10.40	9.68	12.32	7.76	10.65
NO	230	17.43	25.10	15.57	22.80	20.55	27.25	18.93	27.97
NO2	227	13.33	18.24	11.70	15.45	15.78	21.01	15.39	20.07
NOx	231	20.66	26.98	16.47	23.40	22.46	29.09	21.27	30.76
Ozone	227	3.84	5.57	4.37	5.99	4.44	6.03	4.98	6.38
PM10	227	76.71	104.22	68.51	94.57	76.97	107.84	74.46	101.61
PM2.5	228	28.77	43.71	23.14	34.88	28.58	45.09	27.94	40.67
SO2	226	6.81	8.65	7.55	9.34	8.17	9.91	7.86	9.63
Toluene	229	9.67	11.59	9.98	12.00	11.49	13.36	10.86	12.77

Daily forecasts: RMSE comparison
Lower is better.

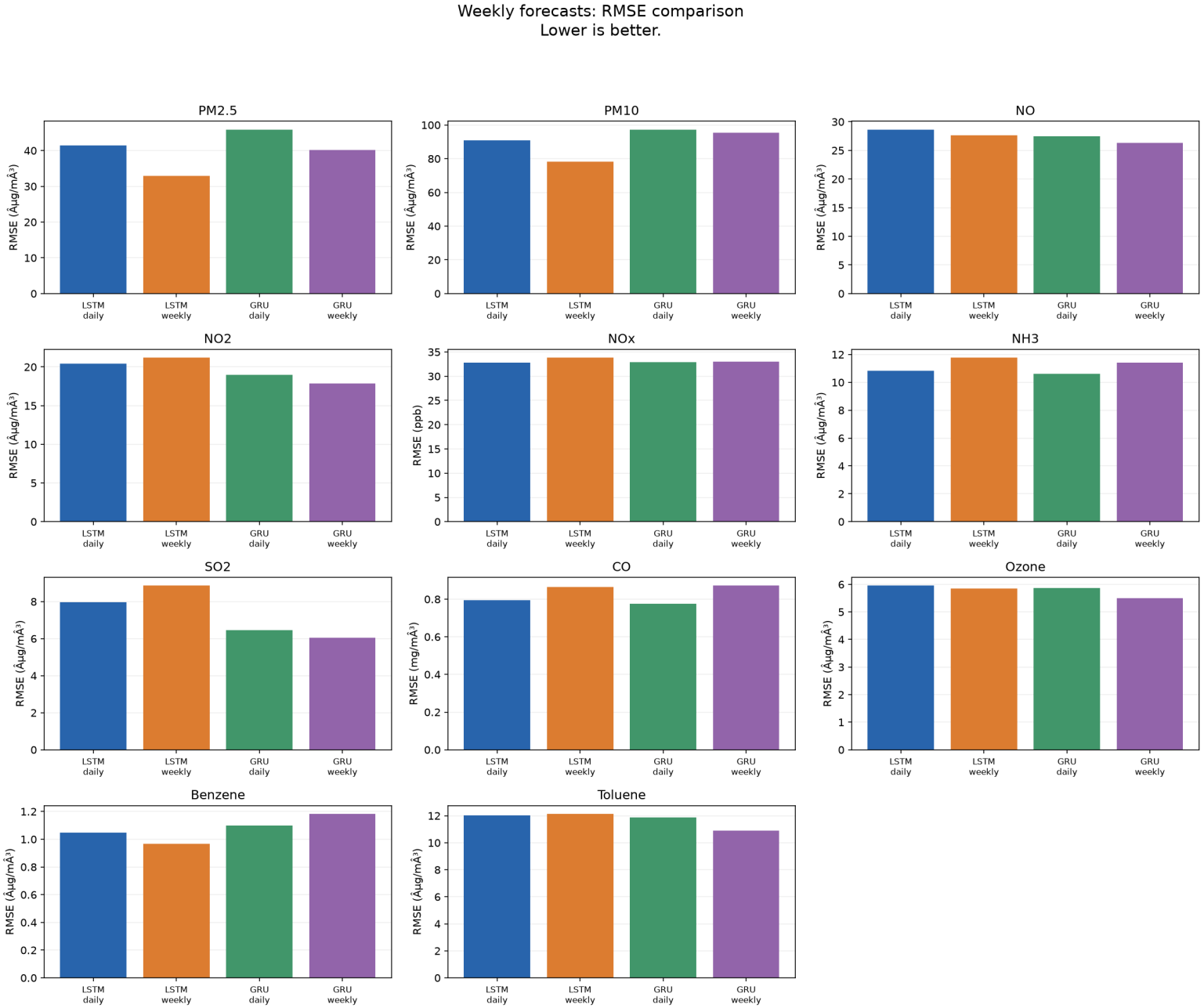


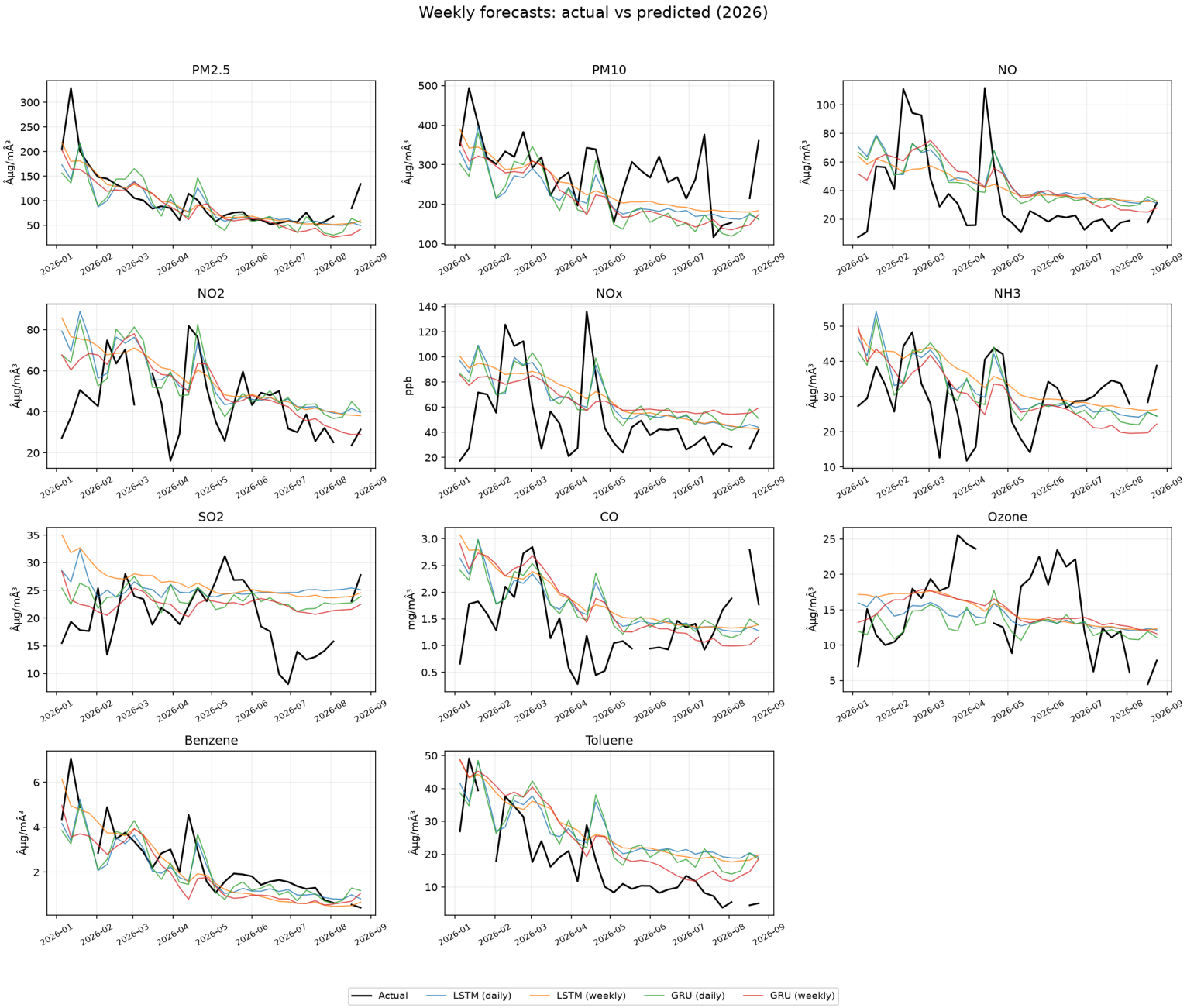

Daily forecasts: actual vs predicted (2026)
PM2.5
PM10
NO
NO2
NOx
NH3
SO2
CO
Ozone
Benzene
Toluene
Actual
LSTM (hourly)
LSTM (daily)
GRU (hourly)
GRU (daily)

Weekly errors

MAE and RMSE in original units. Lower is better. N is the number of scored periods.

Pollutant	N	GRU daily MAE	GRU daily RMSE	GRU weekly MAE	GRU weekly RMSE	LSTM daily MAE	LSTM daily RMSE	LSTM weekly MAE	LSTM weekly RMSE
Benzene	32	0.71	1.10	0.86	1.18	0.66	1.05	0.74	0.97
CO	32	0.60	0.78	0.68	0.87	0.62	0.80	0.69	0.87
NH3	33	8.59	10.62	9.89	11.41	8.89	10.83	9.34	11.78
NO	33	22.19	27.47	21.98	26.35	23.91	28.66	22.67	27.64
NO2	32	14.96	18.98	13.84	17.89	15.85	20.44	16.40	21.21
NOx	33	26.97	32.89	28.92	32.99	26.08	32.84	27.98	33.85
Ozone	32	4.78	5.87	4.68	5.49	5.02	5.95	4.90	5.85
PM10	33	75.73	97.34	75.07	95.51	70.55	91.00	62.80	78.35
PM2.5	33	30.25	45.91	25.54	40.13	24.12	41.43	18.63	33.00
SO2	33	5.30	6.46	4.96	6.04	6.42	7.95	7.27	8.86
Toluene	32	10.88	11.87	9.22	10.91	11.31	12.03	10.98	12.15

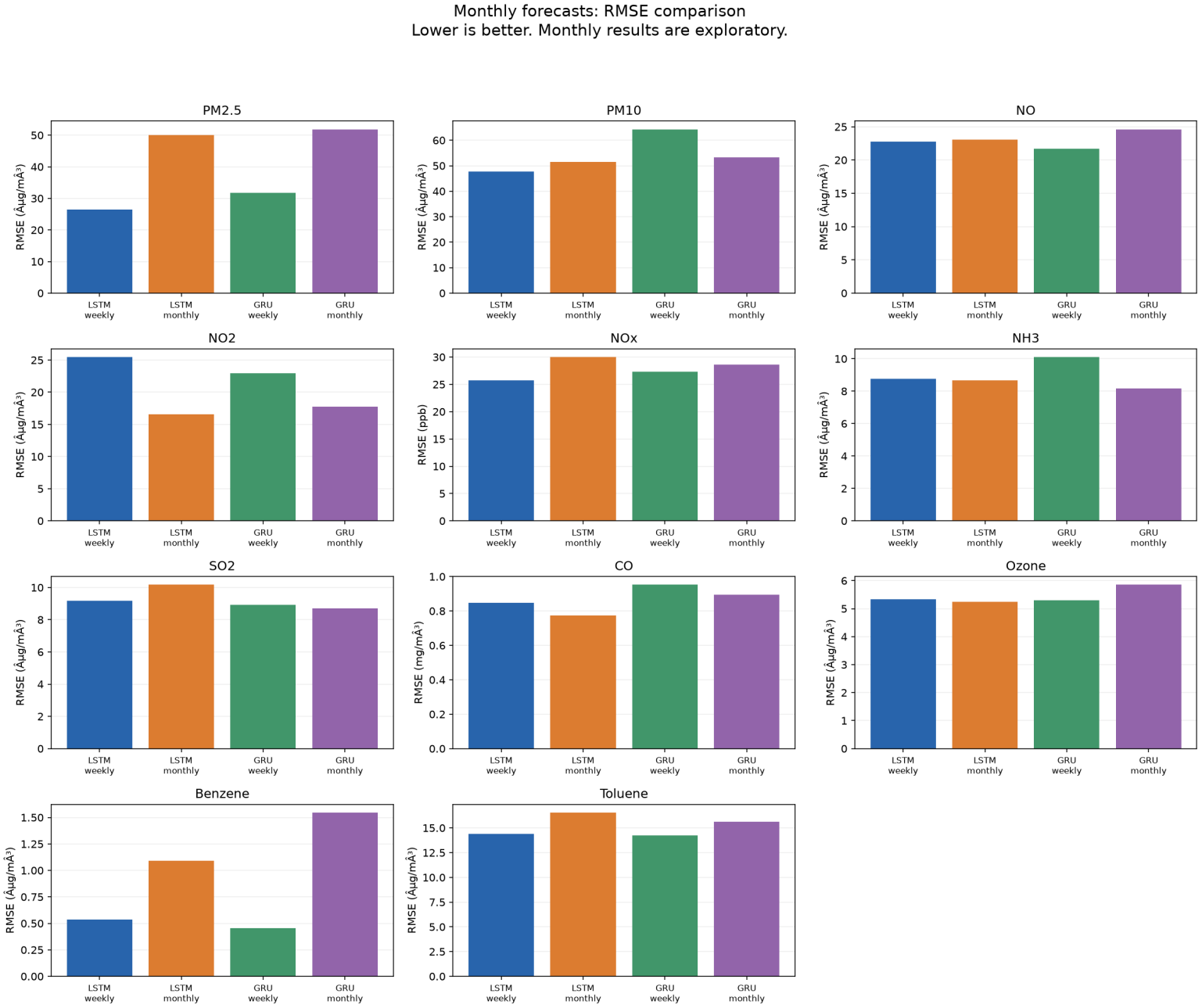

Weekly forecasts: RMSE comparison
Lower is better.
PM2.5
PM10
NO
NO2
NOx
NH3
SO2
CO
Ozone
Benzene
Toluene
RMSE (Âµg/mÂ³)
RMSE (ppb)
RMSE (mg/mÂ³)
LSTM daily
LSTM weekly
GRU daily
GRU weekly

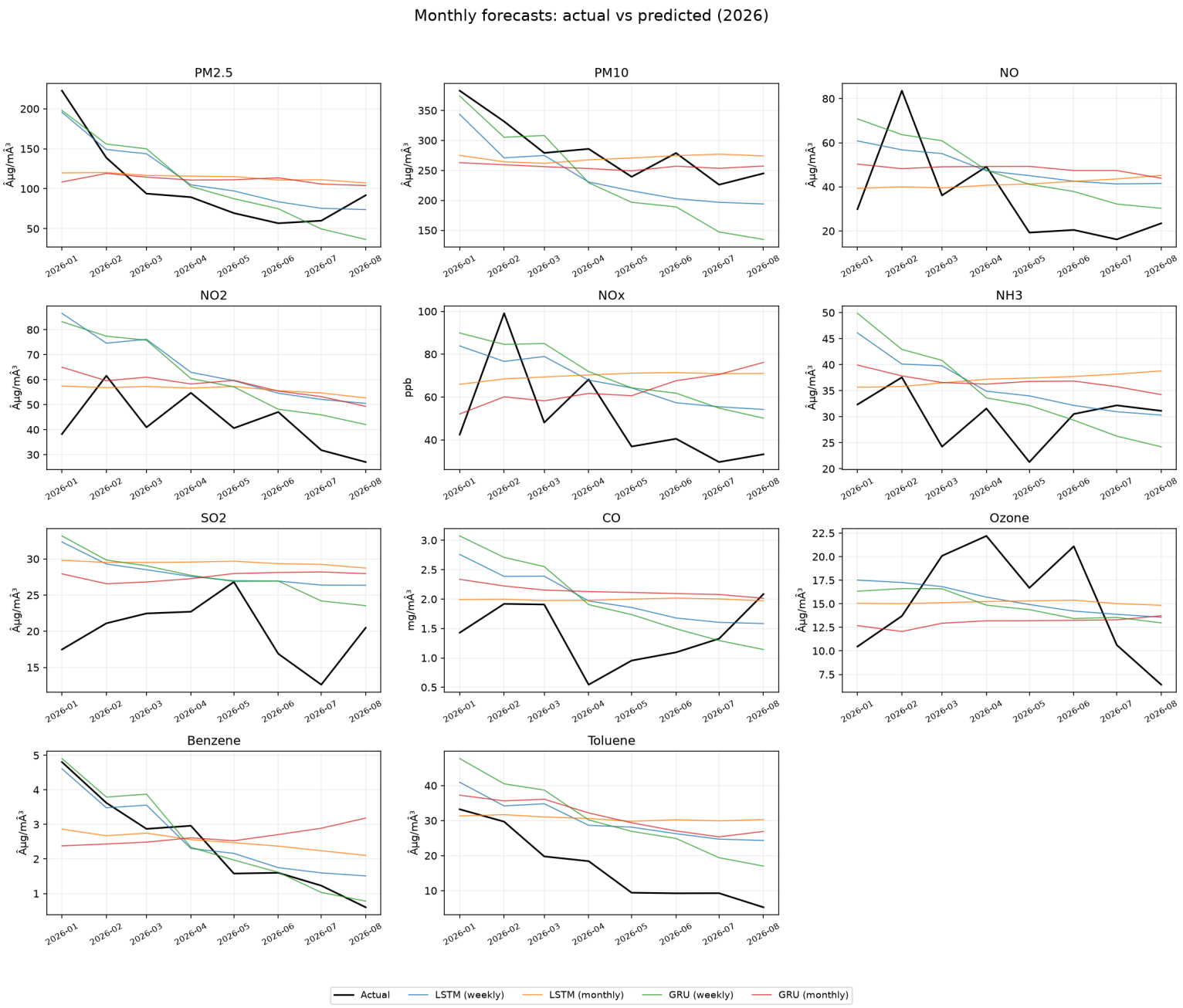

Weekly forecasts: actual vs predicted (2026)
PM2.5
PM10
NO
NO2
NOx
NH3
SO2
CO
Ozone
Benzene
Toluene
Actual
LSTM (daily)
LSTM (weekly)
GRU (daily)
GRU (weekly)

Monthly errors

MAE and RMSE in original units. Lower is better. N is the number of scored periods.

Pollutant	N	GRU monthly MAE	GRU monthly RMSE	GRU weekly MAE	GRU weekly RMSE	LSTM monthly MAE	LSTM monthly RMSE	LSTM weekly MAE	LSTM weekly RMSE
Benzene	8	1.33	1.55	0.33	0.46	0.95	1.09	0.46	0.53
CO	8	0.75	0.89	0.82	0.95	0.61	0.77	0.75	0.85
NH3	8	6.68	8.17	8.30	10.11	7.50	8.68	6.45	8.76
NO	8	22.17	24.62	18.67	21.71	19.74	23.08	21.21	22.78
NO2	8	15.44	17.76	18.51	22.95	14.49	16.56	21.88	25.47
NOx	8	24.99	28.66	24.15	27.28	27.71	30.01	23.24	25.75
Ozone	8	5.17	5.85	4.88	5.30	4.72	5.25	4.93	5.33
PM10	8	39.97	53.33	55.32	64.31	40.88	51.59	42.49	47.63
PM2.5	8	41.55	51.88	26.64	31.71	42.10	49.99	23.84	26.50
SO2	8	7.54	8.70	7.60	8.91	9.36	10.17	7.98	9.18
Toluene	8	14.45	15.62	13.89	14.22	14.31	16.56	13.47	14.38


Monthly forecasts: RMSE comparison
Lower is better. Monthly results are exploratory.
PM2.5
PM10
NO
NO2
NOx
NH3
SO2
CO
Ozone
Benzene
Toluene
RMSE (Âµg/mÂ³)
RMSE (ppb)
RMSE (mg/mÂ³)
LSTM weekly
LSTM monthly
GRU weekly
GRU monthly


Monthly forecasts: actual vs predicted (2026)
PM2.5
PM10
NO
NO2
NOx
NH3
SO2
CO
Ozone
Benzene
Toluene
Âµg/mÂ³
ppb
mg/mÂ³
2026-01
2026-02
2026-03
2026-04
2026-05
2026-06
2026-07
2026-08
Actual
LSTM (weekly)
LSTM (monthly)
GRU (weekly)
GRU (monthly)